

RMinte

RM-01

Hardware Security

Network access · Device binding · Tamper protection

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RM-01 hardware security and supporting management functions

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About this document

The main text describes the security measures and their purpose. Shaded notes explain technical concepts and the scope of protection. Use the contents or PDF bookmarks to navigate between sections.

1. Security Overview

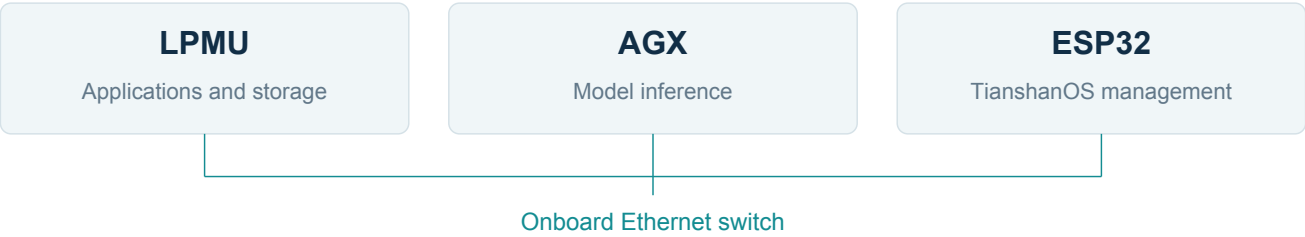
RM-01 incorporates network access controls, storage encryption, device binding, and physical tamper protection into its hardware design. These measures protect the supplied software environment and restrict unauthorized access, transfer, and use.

The design addresses five areas:

- **Defined network access boundaries:** Connecting a device through C1 does not, by itself, expose internal services to the entire office network.
- **Software bound to its designated unit:** Storage encryption and card-to-unit binding prevent the supplied environment from being reused on another unit by moving the card.
- **Compute modules bound to their original carrier board:** Both AGX and LPMU have boot-level binding that prevents unauthorized use on another board.
- **Physical tamper protection:** Custom screws make the enclosure harder to open. Opening it interrupts RTC backup power and prevents the unit from booting.
- **Device identity management:** TianshanOS manages keys, certificates, and connection identities to support authenticated device administration.

This document covers RM-01 hardware and its supporting embedded system. It complements the software supplier's application security documentation. Business accounts, application permissions, and business data management are covered in that documentation.

2. Hardware Architecture and Security Roles



RM-01 consists of three main units for application computing, model inference, and device management, connected through an onboard Ethernet switch. Each unit has a defined role and hosts the corresponding workloads or management functions.

Hardware unit	Primary role	Security role
LPMU application server	An x86 platform that runs the supplier's software and has its own storage.	Hosts business applications. Binding to the original carrier board restricts use of the module in another unit.
AGX inference server	An NVIDIA computing platform running Ubuntu Server and the inference engine.	Provides model inference. Storage-card and module binding protect the inference environment.
ESP32 management unit	Runs TianshanOS for device management, USB switching, and key and certificate management.	Controls USB connection targets and provides device identity management.

This division of responsibilities places the supplier's application storage on LPMU and the operating system and inference storage on AGX, with defined boundaries for operation and management.

Technical note: Compute modules and system software

x86 is the processor architecture used by LPMU. Ubuntu Server is the server operating system used by AGX. The inference engine is software that runs on the operating system and uses AI models to perform computations.

The onboard switch connects the internal devices. Network connectivity and access permissions are separate concerns; service permissions must be configured as part of the deployment.

3. Network Access and USB Controls

3.1 Internal Network Access through C1

C1 connects a computer, tablet, phone, or other endpoint to RM-01. The endpoint detects a USB network interface and joins the RM-01 internal subnet, where it can access the application and management services made available to it.

C1 brings the endpoint onto the RM-01 internal network; it does not automatically place RM-01 on the endpoint's existing office LAN.

For example, a computer may connect to the office network over Wi-Fi and to RM-01 through C1. Unless bridging, forwarding, or another external access path is enabled, other devices on the office network do not gain access to RM-01 simply because C1 is connected.

This design supports local, direct access and reduces the risk of exposing internal services to the office network without realizing it.

C1 is an entry point to the RM-01 internal network and should connect only to trusted, managed endpoints. Connection sharing, bridging, proxies, and port forwarding can change the access boundary and should be configured by an administrator.

Technical note: Subnets and access boundaries

A subnet is a group of networked devices with a defined address range. A computer can connect to both the office network and the RM-01 subnet. Communication between those networks depends on whether a bridge, route, or other forwarding path exists.

C1 does not provide one-way data transfer or an air gap. Its role is to establish a defined access path without exposing services to a wider network merely by connecting a cable.

3.2 Connection Target Control for C3

C3 is the shared USB port at the top of the enclosure. TianshanOS can switch its connection target to ESP32, AGX, or LPMU.

When a direct connection to a compute module is not needed, C3 can be assigned to ESP32 so that it no longer connects to AGX or LPMU. ESP32 does not provide an SSH server login interface. This setting removes an unnecessary direct path to the compute modules through C3.

Switching the target changes only the C3 connection. It does not affect an existing C1 network connection or disable services reachable through other network paths.

Technical note: USB switching and management interfaces

An internal hardware multiplexer switches C3 between modules. The port connects to the selected module at any given time.

SSH is an encrypted protocol for logging in to a device and running commands over a network. A serial interface is a separate management channel used for configuration, logs, and maintenance. C3 may still provide serial access when assigned to ESP32, so the absence of an SSH login interface does not mean the port is disabled.

3.3 Controlled Access from the LAN

To share RM-01 services on a LAN, C3 can be assigned to LPMU and connected to a switch or router through an Ethernet adapter. TianshanOS can then initiate the associated network configuration.

This deployment mode extends the access boundary. An administrator should define which services, devices, and connection paths are needed, without exposing unnecessary management interfaces.

Technical note: Routing and port forwarding

Routing connects different networks. Port forwarding directs connections received at a network entry point to an internal service.

Port forwarding makes the corresponding internal service reachable from a wider network. LAN sharing must therefore be managed as a separate deployment configuration; the access boundary for a local C1 connection no longer applies.

4. Storage Encryption and Software Protection

4.1 Encryption and Card-to-Unit Binding

RM-01 uses a CFexpress Type B card to hold the AGX Ubuntu Server installation, the inference engine, and the supplier's software for delivery. These components are encrypted, and the card is bound to a designated RM-01 unit.

The environment supplied on a card assigned to unit A cannot run on unit B.

This mechanism prevents reuse of the complete system and software by transferring the card to another unit. It protects the supplier's licensing rights and keeps the supplied environment tied to its designated hardware.

A card reader can recognize and read the card. Reading its contents does not remove the device binding or enable the original system and software to run on another unit.

Technical note: Encryption and runtime binding

Encryption protects content; device binding determines which unit can run it. They serve different purposes and work together to protect the supplied environment.

Binding a card to one unit restricts its use. It does not mean a reader cannot read the card, or that every partition and file on it is unreadable.

4.2 Separate Delivery and Application Storage

The supplier's software is delivered on the CFexpress Type B card. When the software runs, RM-01 transfers it to LPMU's own storage and removes it from the card once the transfer is complete. LPMU then hosts the application.

This process separates the delivery medium from the storage used to run the application, reducing the exposure associated with leaving the supplier's software on a removable card. Removing the card after the transfer does not provide access to the application environment stored on LPMU.

Technical note: File handling after transfer

Removal from the card describes a change in where the software is stored. It does not mean the files have been securely erased beyond recovery.

File removal, encryption, and secure erasure are different mechanisms. Protection of applications and data on LPMU's storage depends on the software security plan and the delivered configuration.

4.3 Inference Processing and Business Records

The AGX inference engine receives computation requests and returns results. In this deployment, the CFexpress Type B card is not used as persistent business storage for requests or inference results.

Reading the operating system or inference engine files does not provide access to historical business records or application data stored on LPMU.

Whether conversations, files, or business results are retained, where they are stored, and who may access them are defined by the supplier's application configuration and software security plan.

Technical note: Persistent and temporary data

Persistence means writing data to storage so that it remains available after a program exits or the device loses power.

Model files, temporary computation data, runtime logs, and business records are different types of data. The statement that the inference service does not retain business records is not a promise to erase all model files, caches, or logs.

5. Compute Module Binding

5.1 AGX Boot Binding

AGX uses a device-specific ID and a custom BIOS configuration and is bound to its original carrier board.

Transferring AGX to another board, including an identical board in another RM-01, prevents the system from booting. Matching connectors and board models do not allow the original inference environment to be restored by moving the module.

5.2 LPMU Boot Binding

LPMU uses a custom BIOS configuration and is also bound to its original carrier board.

The system cannot boot if LPMU is transferred to another development board or an identical board in another RM-01. This prevents the original application environment from being used on another board by moving the entire application compute module.

AGX and LPMU binding is enforced at system boot. It protects startup of the computing environment, rather than relying solely on a license check within an individual application.

Technical note: Device identity and boot firmware

A device ID distinguishes one device from another. BIOS is low-level firmware responsible for hardware initialization and system startup; this document uses the product's term for its custom boot firmware.

Connector compatibility establishes a physical connection, not permission to boot. Module binding restricts system startup and use on other carrier boards; it does not establish that every storage medium is unreadable on its own.

6. Physical Tamper and Boot Protection

6.1 Custom Enclosure Protection

The RM-01 structural enclosure is secured with two screws designed exclusively for RM-01. Their design makes disassembly more difficult and raises the barrier to unauthorized access to the internal board and compute modules.

The screws and internal boot protection work together: the screws restrict physical access, while boot protection prevents continued operation after the structural enclosure has been opened.

6.2 Boot Protection Triggered by Enclosure Opening

The RM-01 enclosure design links opening the structural enclosure to the board's RTC backup power. Opening it interrupts RTC battery power and triggers hardware-level protection, preventing the board from operating and the unit from booting.

This design prevents continued use after unauthorized disassembly and protects the original hardware assembly and operating environment.

The manufacturer can restore operation after this protection has been triggered. Internal repairs and recovery should be handled through the manufacturer's service process.

Technical note: RTC and boot protection

RTC stands for Real-Time Clock. The RTC battery supplies backup power to the clock and related circuitry when the main power is off.

RM-01 links this backup power state to its custom boot protection. The startup restriction following loss of RTC power is part of the RM-01 design, not a feature common to all computers.

Opening the enclosure here means disassembling the structural enclosure. It does not include opening the top magnetic cover to access C3 as described in the operating instructions. Triggering boot protection does not automatically erase business data.

6.3 Long-Term Backup Power and Manufacturer Recovery

RM-01 uses a high-quality RTC battery and a low-power backup design to provide at least five years of RTC power, reducing the impact of battery depletion on normal use.

Battery depletion also triggers boot protection. Battery maintenance and enclosure servicing should therefore be handled by the manufacturer to avoid an unintended interruption of backup power.

Boot protection does not mean the unit is permanently unusable. The manufacturer can address the battery and restore startup as part of the service process.

7. TianshanOS Identity and Credential Management

TianshanOS runs on ESP32 and manages keys, certificates, and connection identities. These functions support identity verification for device administration and management of the associated credentials.

7.1 Key Pair Generation and Management

TianshanOS can generate and manage RSA and elliptic-curve key pairs on the device for defined purposes. Keys can authenticate device connections and reduce the need to enter passwords during routine administration.

Private-key export is disabled by default. Only keys marked as exportable when created can be exported through the export function.

Technical note: Public and private keys

A public key is shared with the party verifying an identity, while the holder keeps the private key. During authentication, the holder performs a cryptographic operation using the private key, and the other party verifies it using the public key without receiving the private key.

Disabling export by default is a key management policy, not a guarantee that private keys cannot be extracted under every form of physical attack.

7.2 Public-Key Authorization and Revocation

TianshanOS can deploy public keys to target servers for SSH key authentication and revoke keys that have been deployed.

When an administrative connection is retired or its authorization changes, the corresponding public key can be removed to end its authorization for future logins.

Technical note: Scope of revocation

Revoking a public key removes the authentication permission associated with that key. It does not remove other accounts, passwords, or authorized keys on the target server. Broader permission changes should be coordinated with account management.

7.3 Connection Identity Verification

TianshanOS records and verifies SSH host fingerprints. If a previously recorded host's fingerprint changes, it rejects the connection by default until the change is explicitly accepted.

This identifies changes in the connection target's identity, helping administrators investigate device replacement, system reinstallation, or impersonation instead of relying on the network address alone.

Technical note: Host fingerprints

A host fingerprint is a digest of a server's public key, used to compare the current connection target with the recorded identity.

A trusted record must be established on first connection. Any later change requires investigation. A changed fingerprint does not necessarily indicate an attack, but it should not be accepted without verification.

7.4 Device Certificate Management

TianshanOS can generate device keys and certificate signing requests, request certificates from a certificate service, and install and manage device certificates and CA certificate chains. These functions support certificate-based identity authentication.

Technical note: Digital certificates and CAs

A digital certificate associates an identity with a public key and is issued by a certificate authority (CA). A certificate signing request submits the public key and identity information without submitting the private key.

The services covered by certificate authentication depend on their configuration. Certificate management capabilities do not mean that every interface uses certificate authentication.

8. Operation and Maintenance Requirements

To maintain the security state established at delivery, follow these requirements for device connections, network access, and internal maintenance:

Area	Requirement
Endpoint access	Connect trusted, managed endpoints to C1. Keep unmanaged devices off the internal network.
USB configuration	Set the C3 target to match its intended use and remove unnecessary direct paths to compute modules.
Service sharing	Have an administrator configure LAN access and port forwarding, with a defined scope of exposed services.
Card and module replacement	Confirm device binding and recovery arrangements before replacement. Identical parts are not necessarily interchangeable between units.
Internal servicing	Have the manufacturer service the structural enclosure, board, and RTC battery to avoid triggering boot protection through unauthorized disassembly.
Credential management	Define the purpose, custody, and retirement process for keys and certificates. Revoke authorization that is no longer needed.

These requirements help preserve network access boundaries, device binding, and physical protection, and provide a basis for fault recovery and long-term maintenance.